Hongrui Liu

Toronto, Ontario, Canada | hongrui.liu@mail.utoronto.ca

Portfolio | GitHub | LinkedIn

EDUCATION

Master of Science, Institute of Medical Science

University of Toronto / CAMH | September 2024 - September 2026 | Degree awarded

Supervisor: Daniel Felsky, PhD, Whole Person & Population Modelling Lab. Research in machine learning, lifestyle-associated microbiome profiles, and cognition.

Honours Bachelor of Science

University of Toronto | September 2018 - June 2022

Double major in Mathematics and Human Biology. GPA: 3.97/4.0.

RESEARCH EXPERIENCE

MSc Research - ML-LAMP

Whole Person & Population Modelling Lab, CAMH / University of Toronto | 2024 - 2026

- Developed a workflow to predict microbiome components from lifestyle and anthropometric variables in the American Gut Project, then projected model-derived scores into UK Biobank for association testing.
- Used Python and R for data harmonization, missing-data handling, log transformation, PCA, random forest modelling, nested cross-validation, model serialization, and reproducible figures.
- Interpreted predictions with SHAP and component loadings; examined associations with cognition, gastrointestinal outcomes, and dementia-related phenotypes.
- Communicated limitations of model-derived proxies, cross-cohort transfer, and observational associations through an MSc thesis and research poster.

Computational Neuroscience - Neurophysiology Simulation

PSL 432 course project, University of Toronto | January - April 2022 | Prof. Douglas Tweed

- Simulated neural control of arm movement and numeral recognition using deep-learning and reinforcement-learning models in MATLAB.

MAPK Pathways in Oxidative Stress

HMB 312 course project, University of Toronto | January - April 2022

- Investigated MAPK responses to oxidative stress in HeLa cells using RT-qPCR and RNA-seq datasets; focused on MAPK1 and p38 signalling and presented a poster and oral presentation.

RESEARCH OUTPUT

Liu H. *ML-LAMP: Machine-Learning-Derived Lifestyle-Associated Microbiome Profiles for Gut-Brain Association Testing.* MSc thesis, University of Toronto, 2026. Research poster with Daniel Felsky.

TECHNICAL SKILLS

Programming: Python (NumPy, pandas, scikit-learn, PyTorch, TensorFlow), R (tidyverse, ggplot2, SparkR), MATLAB.

Methods: random forests, regularized regression, PCA, nested cross-validation, SHAP interpretation, data harmonization, and reproducible analysis.

Tools: Jupyter, RStudio, Git/GitHub, Linux/WSL, high-performance computing, Spark/Delta workflows, and Microsoft Office.

TEACHING AND PROFESSIONAL EXPERIENCE

High School Mathematics Tutor

Mathnasium, Toronto | January 2024 - Present

- Teach high-school mathematics, strengthen problem-solving skills, and support preparation for college and university entrance assessments.
- Adapt explanations to students with different levels of mathematical confidence and learning needs.

Ophthalmology Medical Technician

Toronto Eye Specialists and Surgeons | June 2022 - September 2024

- Performed OCT imaging, visual-field testing, dilation exams, and pre-consultation assessments.
- Collaborated with ophthalmologists, technicians, and front-desk staff on patient flow, documentation, and clinical communication; explained procedures to patients from diverse backgrounds.

Customer Service Representative - Hot Meal Service

Loblaws, Toronto | September 2021 - January 2022

- Served customers, handled orders, and resolved issues in a fast-paced environment.

LEADERSHIP AND COMMUNITY

Saturday Program - Leadership and Research Involvement

University of Toronto Medical Society | Leadership through June 2026; research involvement ongoing

- As co-director, led a mentorship and tutoring initiative supporting marginalized high-school students interested in health-care careers.
- Coordinated volunteer recruitment, training, logistics, weekly programming, ceremonies, workshops, and program evaluation processes.
- Developed a Python-based tutor-tutee matching tool using subject, availability, and preference constraints.
- Continue research involvement with the program following completion of leadership responsibilities.

Hospice Companion

Hospice Toronto | June 2022 - Present

- Provide emotional and practical support to patients and families through active listening and companionship.

Knowledge Translator

Meta-Lifeline Food Bank | June 2022 - February 2023

- Synthesized evidence on social determinants of health affecting food-insecure populations and translated health information for community stakeholders.

HONOURS AND AWARDS

University of Toronto Fellowship - Institute of Medical Science
September 2024 | $10,000.

Institute of Medical Science Entrance Scholarship
September 2024 | $2,500.

Victoria College Scholarships
2019 - 2021 | 5T3, J.C. McQueen, and Dale Anne Freed scholarships. Combined value: approximately $3,000.